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Veracity Radius - Capturing the Locality of Distributed Computations

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Abstract

This paper focuses on *local* computations of distributed aggregation problems on fixed graphs. We define a new metric on problem instances, *Veracity Radius (VR)*, which captures the inherent possibility to compute them locally. We prove that VR yields a tight lower bound on output-stabilization time, i.e., the time until all nodes fix their outputs, as well as a lower bound on quiescence time. We present an efficient aggregation algorithm, I-LEAG, which reaches both output stabilization and quiescence within a time that is proportional to the VR of the problem instance, and is also efficient in terms of per-node communication and memory. We empirically show that the VR metric also effectively captures the performance of previously suggested efficient aggregation protocols, and that I-LEAG significantly outperforms these protocols in several respects.

Keywords: aggregation, locality, distributed computation.

1 Introduction

Background We consider distributed computation by a large set of network nodes, which must collaboratively compute and come to know the value of a global *aggregation function* [23, 15] of their individual inputs. For example, testing whether the number of sensors reporting a problem exceeds a certain threshold (majority vote with threshold), determining the average processor load in a computing grid, etc. Such networks may comprise millions of nodes, and communication bandwidth as well as energy resources may be limited, requiring good scalability. An appealing approach for achieving scalability is using *local computations*, whereby the computation time and the amount of per-node communication are independent of the size of the network.

Unfortunately, virtually all interesting aggregation functions, e.g., majority, AND/OR, average, and minimum, cannot always be computed locally – they have instances that require some nodes to indirectly communicate with a fixed fraction of the graph nodes in order to deduce the correct result. Yet, they also have instances that intuitively should require little if any computation (e.g., when no sensor reports a problem). Indeed, recent work presents efficient solutions to interesting aggregation problems, and empirically shows that these solutions *usually* perform local computations [3, 22].

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